

Functional Description

Electronic Equipment

Systembus Coupling Unit SCU 1

E532405/02E



Power. Passion. Partnership.

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1 Safety

1.1 Important provisions for all products

Nameplate

The product is identified by nameplate, model designation or serial number and must match with the information on the title page of this manual.

Nameplate, model designation or serial number can be found on the product.

All EU-certified engines delivered by MTU come with a second nameplate. When operating the machine in the EU: The second nameplate must be affixed in a prominent position as described in the accompanying specifications.

General information

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Modifications or conversions
- Noncompliance with the safety instructions and warning notices

Correct use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/MTU contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (including engine control/parameters)
- In compliance with all safety regulations and in adherence with all warning notices in this manual
- With maintenance work performed in accordance with the (→ Maintenance Schedule) throughout the useful life of the product
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair
- By contracting only workshops authorized by the manufacturer to carry out repair and overhaul

Any other use is considered improper use. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

Emission regulations and emission labels

Responsibility for compliance with emission regulations

Modification or removal of any mechanical/electronic components or the installation of additional components including the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission control units/systems may only be maintained, exchanged or repaired if the components used for this purpose are approved by the manufacturer.

Noncompliance with these regulations will invalidate the design type approval issued by the emissions regulation authorities. The manufacturer does not accept any liability for violations of the emission regulations. The maintenance schedules of the manufacturer must be observed over the entire life cycle of the product.

Replacing components with emission labels

Emission labels are attached to all MTU engines. These must remain on the engine throughout its operational life.

Engines used exclusively in land-based, military applications other than by US government agencies are excepted from this proviso.

Please note the following when replacing components with emission labels:

- Appropriate emission labels must be affixed on spare parts.
- Emission labels may not be transferred from old components to new ones.
- Emission labels on old components must be removed and destroyed.

1.2 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair and transport personnel at all times.

Personnel must receive instruction on product handling and maintenance on the basis of this manual with a special emphasis on safety requirements and warnings.

This is particularly important in the case of personnel who only occasionally perform work on or around the product. This personnel must be instructed repeatedly.

Personnel requirements

All work on the product shall be carried out by trained and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel specialized in mechanical and plant engineering

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair and transport.

Working clothes and personal protective equipment

Wear proper protective clothing for all work.

When working, always wear the necessary personal protective equipment (e.g. safety footwear, ear protectors, protective gloves, goggles, breathing protection). Observe the information on personal protective equipment in the respective activity description.

1.3 Safety regulations for startup and operation

Safety requirements for startup

Install the product correctly and carry out acceptance in accordance with the manufacturer's specifications before putting the product into service. All necessary approvals must be granted by the relevant authorities and all requirements for initial startup must be fulfilled.

When putting the product into operation, always ensure that

- All personnel is clear of the danger zone surrounding moving parts of the machine.
Electrically-actuated linkages may be set in motion when the Engine Control Unit is switched on.
- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating machine components.
- All safety equipment is in place.
- No persons wearing pacemakers or any other technical body aids are present.
- The service room is adequately ventilated.
- Keep clear of the service room during the first operating hours. Hazardous gases may occur as a result of the combustion of paints or oils.
- The exhaust system is leak-tight and that the gases are vented to atmosphere.
- Protect battery terminals, generator terminals or cables against accidental contact.
- Check that all connections have been correctly allocated (e.g. +/- polarity, direction).

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems are working properly.

Safety requirements for operation

The operator must be familiar with the controls and displays.

The operator must be familiar with the consequences of any operations performed.

During operation, the display instruments and monitoring units must be permanently observed with regard to present operating status, violation of limit values and warning or alarm messages.

Malfunctions and emergency stop

The procedures for emergencies, in particular, emergency stop, must be practiced regularly.

The following steps must be taken if a malfunction of the system is detected or reported by the system:

- Inform supervisor(s) in charge.
- Analyze the message.
- Respond to the emergency appropriately, e.g. execute an emergency stop.

Operation

Do not remain in the operating room when the product is running for any longer than absolutely necessary.

Keep a safe distance away from the product if possible. Do not touch the product unless expressly instructed to do so following a written procedure.

Do not inhale the exhaust gases of the product.

The following requirements must be fulfilled before the product is started:

- Wear ear protectors.
- Mop up any leaked or spilled fluids and lubricants immediately or soak up with a suitable binder agent.

Operation of electrical equipment

When electrical equipment is in operation, certain components of these appliances are electrically live.

Observe the safety instructions for these devices.

1.4 Standards for safety notices in the text

DANGER 	In the event of immediate danger. Consequences: Death, serious or permanent injury! • Remedial action.
WARNING 	In the event of a situation involving potential danger. Consequences: Death, serious or permanent injury! • Remedial action.
CAUTION 	In the event of a situation involving potential danger. Consequences: Minor or moderate injuries! • Remedial action.
NOTICE 	In the event of a situation involving potentially adverse effects on the product. Consequences: Material damage! • Remedial action. • Additional product information.

Safety notices

1. This manual with all safety instructions and safety notices must be issued to all personnel involved in operation, maintenance, repair or transportation.
2. The higher level warning notice is used if several hazards apply at the same time. Warnings related to personal injury shall be considered to include a warning of potential damage.

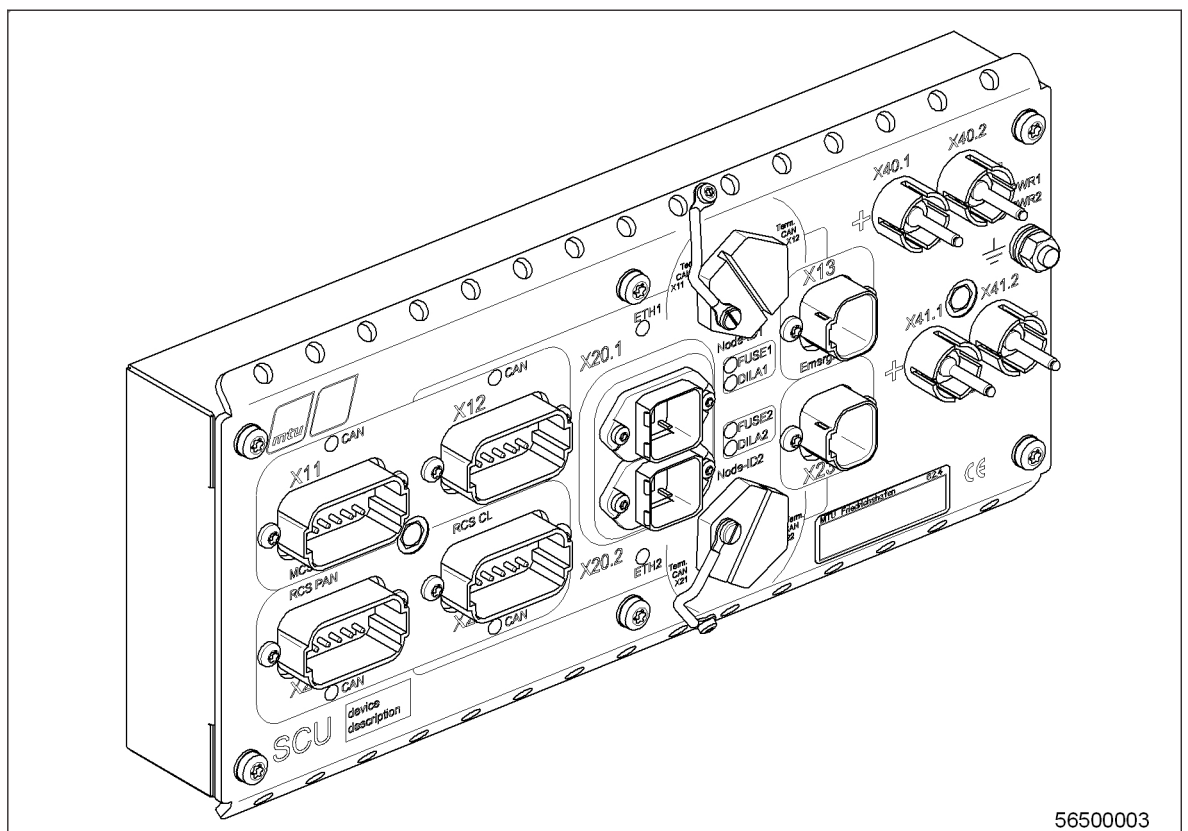
2 Functional Description

2.1 Function of Systembus Coupling Unit SCU 1

Systembus Coupling Unit SCU 1 is a CAN-over-Ethernet (CoE) interface serving to link up the system bus and the CAN bus. The unit features two independent processing units each of which providing two CAN interfaces. Input and output data can be transmitted via the CAN interfaces. The unit processes CAN Layer 2 packages and supports CAN protocols based thereon (e.g. CANopen, SAE J1939, PCS-5).

The two processing units exchange transmitted and received CAN packages with the other stations on the system bus. The two processing units can operate on a redundant power supply.

SCU 1 is typically used to link up control stand components via CAN and to exchange data with the Local Operating Panel LOP in the engine room via the system bus.



SCU 1 has the following tasks:

- Communication interface between system bus and CAN bus
- Connection of main and slave control stands to the system bus, e.g.:
 - Power supply for connected CAN components
 - Transfer of data traffic via system bus.

2.2 Design of Systembus Coupling Unit SCU 1

Mechanical design of SCU 1

